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Liquidity Prediction Model

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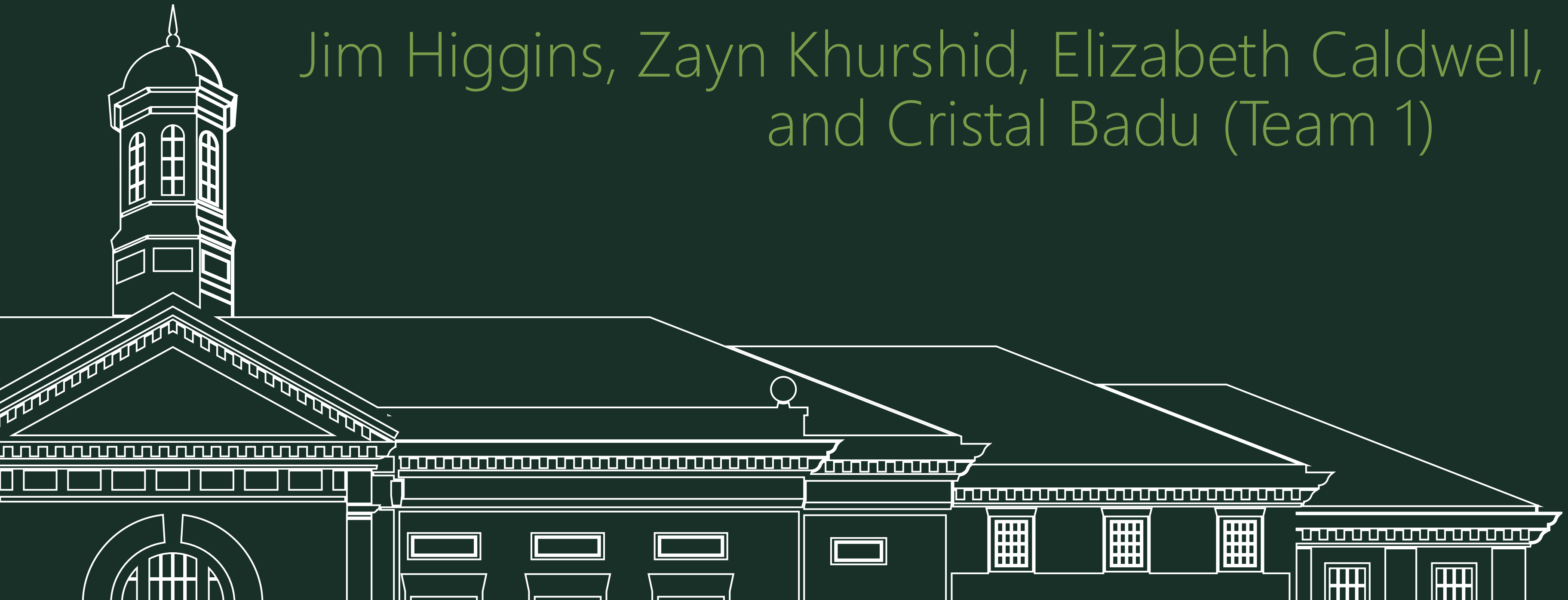


Background & Problem

Project Context & Objective

- Founders often spend 12+ months reaching out to hundreds of investors
- Key challenge:
  Difficult identifying which investors actually have capital available
- Investor activity ≠ investor liquidity

Core Problem & Objective

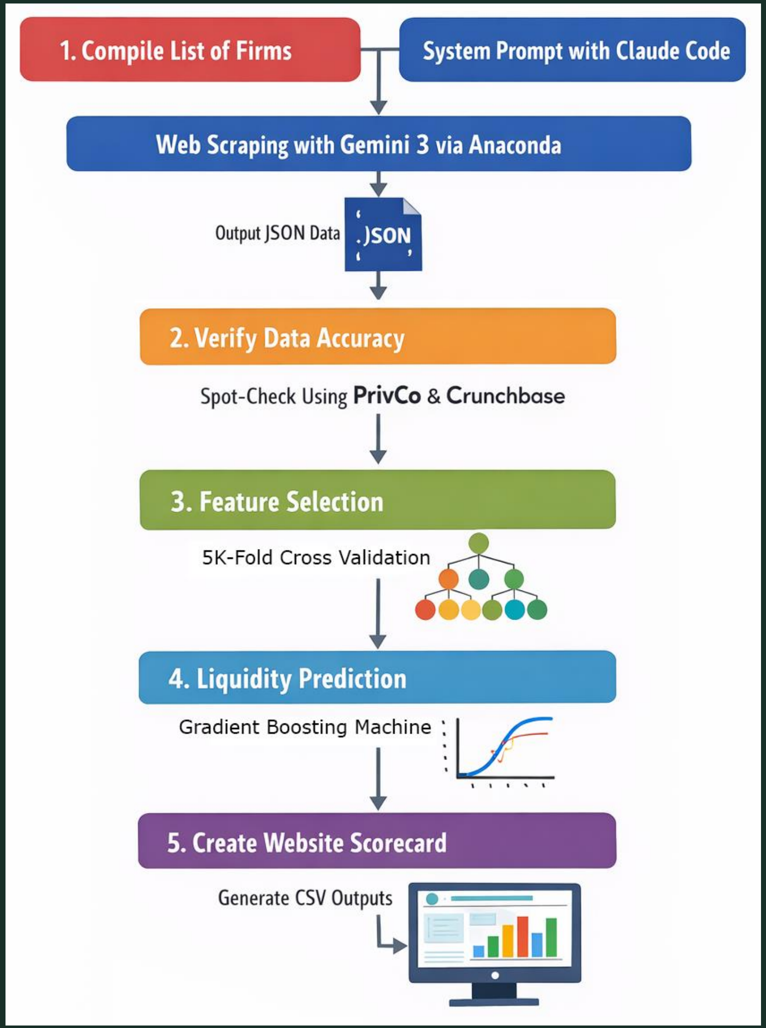
- Many firms appear active but are not currently deploying capital

Solution → Develop a **Liquidity Scorecard**

- **Identify** investors most likely to be actively investing now
- **Prioritize** outreach for founders
- **Reduce** time spent on inactive investors

Methodology Overview

1. Data collection & validation
2. Key indicator selection
3. Model development & evaluation
4. Liquidity prediction & scorecard creation



Data

Data Sources

- Crunchbase & PrivCo
- SEC filings
- Firm websites & press releases
- Credible news sources

Data Pipeline

- AI-driven scraping via Gemini API → JSON output
- Outputs structured for Python modeling
- Scraped data spot-checked against PrivCo and Crunchbase for accuracy

Scope

Data Scope

- Focus: Medtech, Healthtech, Digital Health
- Stage: Seed & Series A
- Timeframe: 2022-2026
- ~100+ VC firms
- Firm level aggregation for consistency
- All public data

Key Indicators

- Average deals per year
- New fund (last 12 months)
- Follow-on activity
- Fund age & size
- Average round size

How To Define Liquidity

Liquidity is not directly observable - we estimate it using:

- Recent investment activity
  - Invested in the last 12 months

- Fund lifecycle stage
  - Current fund is within an active deployment window (~ <5 years)

Liquidity Model Overview

Key Variables (What We Collected)

- Recent Activity
  - Investments in last 12 months (Active Investor)
- Fund Characteristics
  - Fund age (Lifecycle stage)
  - Years since last fund raised
- Investment Behavior
  - Avg deals per year
  - Avg round size ($)
  - % capital deployed
- Portfolio Signals
  - Follow-on investments (# and %)
  - Total exits (2022–present)
  - These variables act as ***proxies for liquidity*** (since liquidity isn't directly observable)

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Model Results & Key Insights

Model Performance Comparison

- Gradient Boosting Machine (GBM) → best overall performance
  - GBM: ~0.78 accuracy, ~0.87 ROC-AUC
    - Highest accuracy and ROC-AUC
- Random Forest → strong performance, higher recall
- Logistic regression → most interpretable

	model	accuracy	precision	recall	f1	roc_auc
1	GBM	0.7755	0.7857	0.8148	0.8000	0.8721
2	Random Forest	0.7551	0.7273	0.8889	0.8000	0.8552
0	Logistic Regression (H2O GLM)	0.7143	0.6757	0.9259	0.7812	0.8485

Key Takeaway

GBM provides the strongest predictive performance

Logistic Regression supports transparency and explainability

Validation

Model evaluated using cross-validation to ensure performance on unseen data

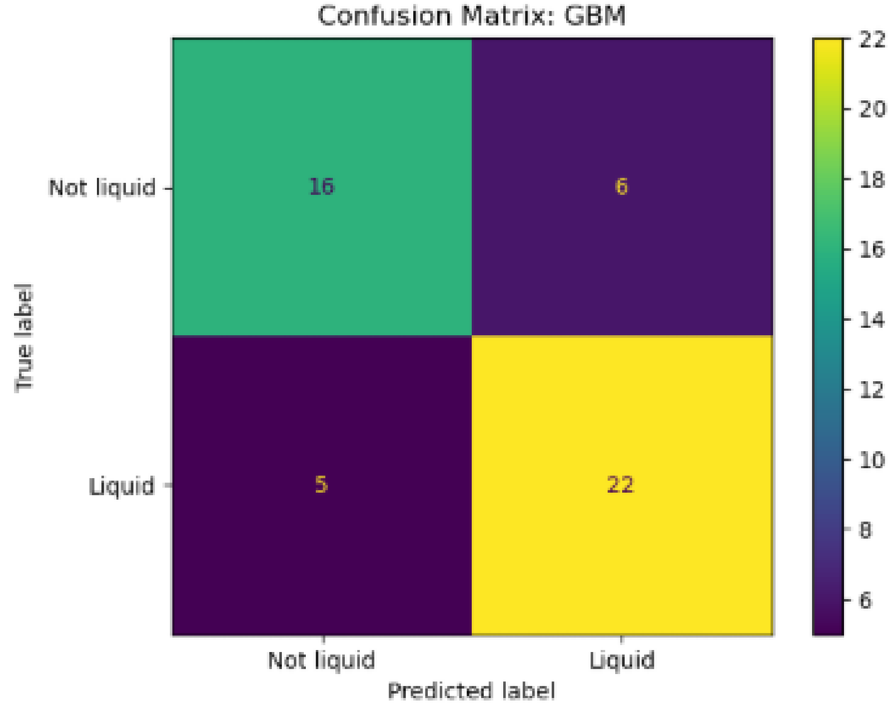
Model Performance Breakdown

Evaluated using cross-validation on full dataset

- 22 Liquid firms correctly identified
- 16 Non-liquid firms correctly identified
- 6 False positives
    - Predicted liquid, but actually not
- 5 False negatives
    - Missed truly liquid firms

*The model is better at identifying **liquid investors** than non-liquid ones*

- High recall = **captures most active investors**
- Some false positives = may include a few inactive firms



	firm	actual_liquid	predicted_probability	predicted_class
48	Sofinnova Partners	1	0.9388	1
33	New Enterprise Associates	1	0.9358	1
10	Andreessen Horowitz	1	0.9059	1
35	F-Prime Capital	1	0.9048	1
47	AlleyCorp	1	0.8990	1
0	5AM Ventures	1	0.8899	1
9	Sequoia Capital	1	0.8896	1
39	Pitango HealthTech	1	0.8541	1
28	Seventure Partners	1	0.8455	1
12	Threshold Ventures	1	0.8407	1

Byond vs GBM

Applied to client-provided dataset to evaluate our performance on the firms in which we have overlap.

- Of the firms which were shared between the datasets
- 22 Liquid firms correctly identified
- 3 Non-liquid firms correctly identified
- 3 False positives
    - Predicted liquid, but actually not
- 1 False negatives
    - Missed truly liquid firm

	CONFUSION MATRIX		
Actual: Not Liquid (0)	3	3	
Actual: Liquid (1)	1	22	
	Predicted: Not Liquid (0)	Predicted: Liquid (1)	

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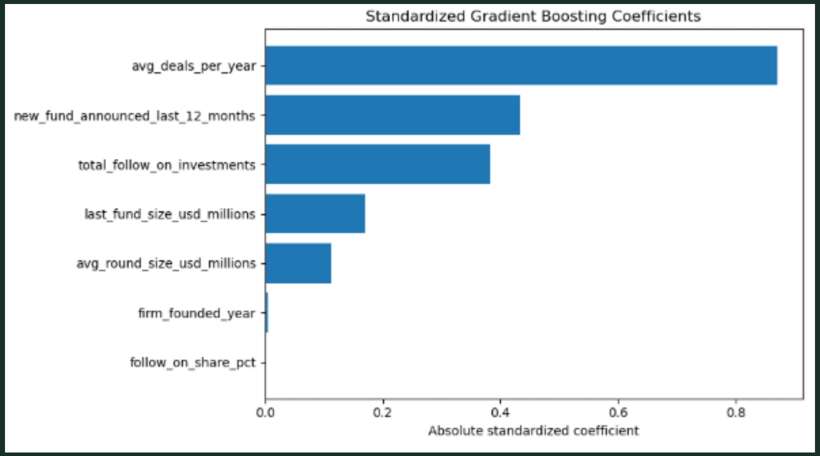
Key Drivers of Liquidity



Two strongest predictors:

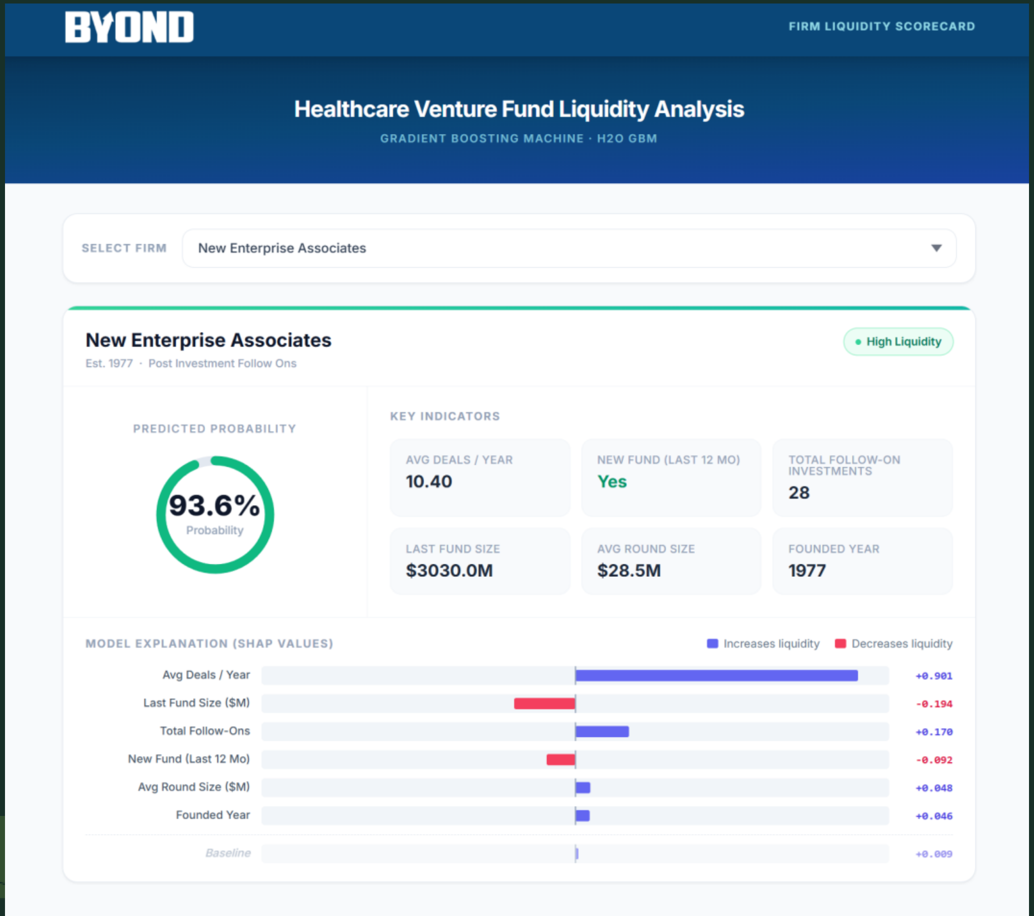
- New fund announced (last 12 months)
- Average deals per year

- Firms with recent capital + consistent deal activity
  - More likely to be actively investing
  - More likely to have available capital

Why This Matters

- Identifies investors actively deploying capital now
- Helps founders prioritize outreach
- Reduces time spent on inactive firms

Scorecard



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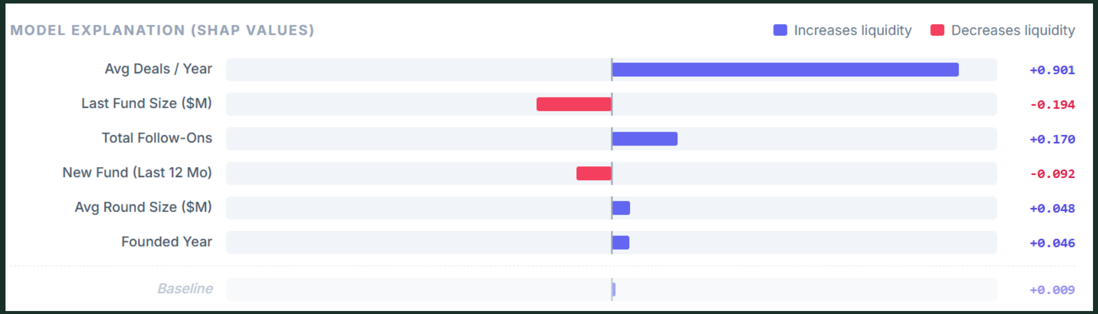
SHAP

- Explains how each indicator affects the prediction
- Breaks the value into positive and negative contributions



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Limitations

- Inherent uncertainty
- Only publicly available sources
- Data is heavily skewed in favor of liquid firms
- Timing
- Limited sample size
- Incomplete data

Looking Forward

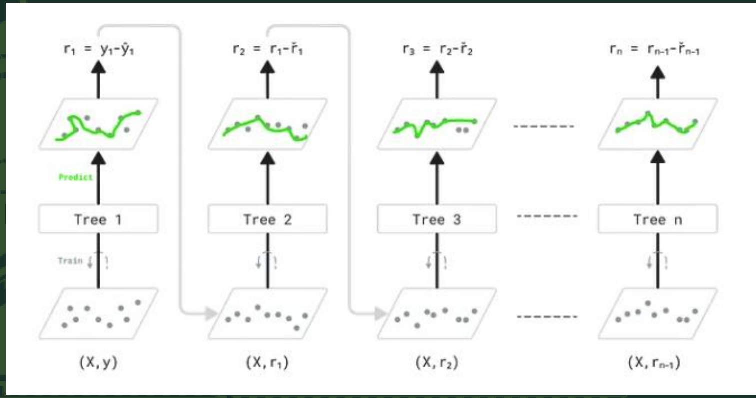
- Expand the dataset
- Explore additional indicators
- Tailor scorecard to clientele
- Fine-tune definition of liquidity
- Craft a system prompt which can access private data

Questions?

What is GBM?

Think of it as a team of simple guessers.

- It's a decision tree
- One makes a rough guess
- The next fixes part of the error
- More trees fix what is still wrong
- The final answer is all those fixes combined



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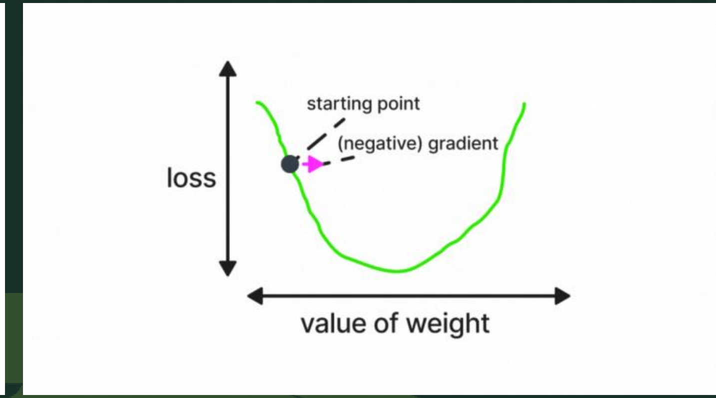

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How it learns

1. **Make a rough first guess**
   A small decision tree gives an initial prediction.
2. **Measure what it got wrong**
   The next tree focuses on the errors and adds a small fix.
3. **Repeat many times**
   Each new tree corrects what the earlier trees still missed.

Final prediction = first guess + many small corrections

Summary Stats

Variable overview

	variable	dtype	non_null_count	missing_count	missing_pct	unique_non_null	example_values
0	pct_capital_deployed	float64	32	161	83.419689	20	70.0 | 65.0 | 46.0
1	total_exits_2022_present	float64	103	90	46.632124	16	4.0 | 0.0 | 2.0
2	visible_down_rounds	float64	103	90	46.632124	3	1.0 | 0.0 | 0.0
3	last_fund_size_usd_millions	float64	160	33	17.098446	102	265.0 | 63.5 | 100.0
4	avg_round_size_usd_millions	float64	163	30	15.544041	132	19.0 | 19.0 | 1.8
5	follow_on_share_pct	float64	187	6	3.108808	102	41.17 | 23.0 | 33.3
6	avg_deals_per_year	float64	189	4	2.072539	74	3.4 | 0.4 | 2.4
7	firm_founded_year	float64	189	4	2.072539	44	2005.0 | 2014.0 | 2017.0
8	currently_raising_from_lps	object	191	2	1.036269	2	False | True | False
9	new_fund_announced_last_12_months	object	191	2	1.036269	2	False | True | False
10	total_follow_on_investments	float64	191	2	1.036269	36	7.0 | 1.0 | 4.0

Numeric summary

	variable	count	missing_count	missing_pct	mean	std	min	p25	median	p75	max	skew
0	firm_founded_year	189	4	2.072539	2009.275132	14.102153	1911.00	2005.00	2014.0	2018.000	2024.00	-2.797071
1	last_fund_size_usd_millions	160	33	17.098446	524.137550	1324.965287	5.00	75.00	172.5	428.750	12500.00	6.342029
2	avg_deals_per_year	189	4	2.072539	4.196138	4.678754	0.00	1.80	2.8	4.250	36.50	3.527685
3	avg_round_size_usd_millions	163	30	15.544041	20.089847	33.898388	0.15	3.20	10.9	19.500	306.00	4.719950
4	total_follow_on_investments	191	2	1.036269	10.073298	18.404086	0.00	2.00	4.0	10.000	150.00	5.069724
5	follow_on_share_pct	187	6	3.108808	40.412620	23.560956	0.00	23.25	38.0	53.565	133.33	0.768819
6	visible_down_rounds	103	90	46.632124	0.067961	0.289087	0.00	0.00	0.0	0.000	2.00	4.647277
7	total_exits_2022_present	103	90	46.632124	4.223301	7.041797	0.00	1.00	2.0	6.000	64.00	6.230047
8	pct_capital_deployed	32	161	83.419689	64.492188	31.263289	3.50	38.75	80.0	90.000	100.00	-0.580414